

What is Population Health Science?

PHS LAUNCH 2026

Activity: Hot Topic in Population Health Science

1

Step 1: Brainstorm a list of current "hot" topics in Population Health Science

2

Step 2: Divide into groups of approximately 5
• **Block Sampling(!)** by field of study

3

Step 3: Choose a topic from the brainstormed list for your group

4

Step 4: Guided group discussion

5

Step 5: Report back to the whole group

Step 1: Making a List of Hot Topics

Possible topics:

- Vaccine hesitancy, health communication and misinformation
- Climate change, environmental justice, and health
- Access and barriers to healthcare
- More suggestions...

Steps 2 and 3: Divide into groups and choose your preferred topic!

Step 4: Group Discussion

Discuss the following within your group:

1. Building on your experiences and norms in your field, what is a research question you would like to explore in this area?
2. What are research tools (concepts or methods) that you might deploy?
3. Working with the diverse perspectives in your group, sketch out an interdisciplinary/transdisciplinary project that builds on your discussion.



Real examples of Population Health Sciences
(PHS 2000A)
in practice

[Former PHS Student Dissertations](#)

What is Population Health Science

PHS 2000A

- Sampling
- Maximum Likelihood Estimation
- Generalized Linear Models
- Survival Analysis
- Measurement
- Multilevel and Longitudinal Models

PHS 2000B

- Causal Inference
- Sensitivity Analysis
- Econometrics
- Missing Data
- Meta-Analysis



Regression – Generalized Linear Models

PFAS Exposure and Cardiometabolic Health in U.S. Women

Susanna Mitro, Epidemiology, 2020

We used multivariable *linear regression* to evaluate the associations of PFAS concentrations with 1-year postpartum weight retention and 3-year postpartum weight change. We used *logistic regression* to estimate odds of substantial 1-year postpartum weight retention (defined as retaining ≥ 5 kg at 1-year postpartum, based on a cutoff commonly used in prior literature as well as prior analyses in this cohort)

Challenges and Opportunities in Epidemiology of a Natural Disaster: Lessons From the 2011 Great East Japan Earthquake and Tsunami

Koichiro Shiba, Social and Behavioral Sciences, 2020

We compared four approaches to estimate the associations between the pre-disaster distance from the coast and depressive symptoms: 1) crude univariate *Poisson regression*, 2) multivariable-adjusted *Poisson regression*, 3) *Poisson regression* weighted by IPTW and IPCW, and 4) the SACE estimation



Survival Analysis

Etiology of Glioma and Glioblastoma

David Cote, Epidemiology, 2019

Tests of linear trend in glioma risk for increasing categories of coffee, tea, and caffeine were evaluated by assigning the median values for each beverage intake category and treating those as a single continuous variable, using *Cox proportional hazards regression*.

Health System Quality and Improvement Strategies in Caring for People With HIV and Cardiovascular Diseases: Studies From Tanzania, Uganda, and China

David Sando, Global Health and Populations, 2019

Cox regression models were used to analyze the association between facility-level quality of care and risk of mortality, adjusting for patient-level risk factors.

Aspirin and Alcohol in Relation to Lethal Prostate Cancer

Mary Downer, Epidemiology, 2018

In both cohorts, regular questionnaires ascertained medical diagnoses, medications, diet, and lifestyle factors. *Cox proportional hazards regression* was used to investigate all associations.



Measurement

BMI and Physical Activity in Relation to Mortality Among Adults With Coronary Heart Disease

Laila Al-Shaar, Nutritional Epidemiology, 2019

Evaluated the performance of a 152-item semiquantitative food frequency questionnaire (SFFQ) as compared to multiple automated self-administered 24-hour dietary recalls (ASA24), two 7-day diet records (7DDR), and several biomarkers in men

Essays on HIV in Rural KwaZulu-Natal, South Africa: Inference From Measurement to Policy

Noah Haber, Global Health and Populations, 2017

We conduct a field experiment to identify whether list randomization can reduce social biases in survey responses related to HIV, using known truths for comparison

Sex, Drugs, and Viral Load: Novel Epidemiologic Methods in HIV Research

Krystal Kantos, Epidemiology, 2019

Psychological instrument validity indices: We also examined BASC-2 validity indices for the participant-reported emotional symptoms index, including the F, L, V, consistency, and response pattern indices. Validity indices alert researchers to the presence of questionable data.



Multilevel and Longitudinal Models

Mass Incarceration and Aggressive Policing: Contextual and Direct Determinants of Population Health

Jaquelyn Jahn, Social and Behavioral Sciences, 2020

We linked county jail incarceration rates to birth certificate data for all births to resident non-Hispanic Black and White U.S. women (N=41,911,094). Using ***multilevel logistic regression models***, we estimated the association between quintiles of county jail incarceration rates and the odds of preterm birth, adjusting for maternal- and county-level covariates and state fixed effects

HIV Risk, Prevention and Testing in Sub-Saharan Africa: Mixed Methods Analysis

Alexander Radunsky, Global Health and Populations, 2019

Using ***longitudinal panel data*** I estimated the risk of HIV among, those who are single and not in conjugal relationships, those who are in longterm unmarried conjugal relationships, and those who are in married conjugal relationships

Cycles of Disadvantage: Eviction and Children's Health in the United States

Gabriel Schwartz, Social and Behavioral Sciences, 2020

Using linear regression and selection weights, they analyze ***longitudinal data*** from a national, urban birth cohort, including only children living in rental housing (N=1,724 for eviction during infancy, 2,126 in early childhood, and 1,979 in middle childhood).